CAPSTONE EVIDENCE NOTE

Title: Review of Suspicious Authentication Activity Across Windows and Linux
Lab: P01-M03-L01 — Windows, Linux and DHCP log analysis
Evidence date: March 14, 2026, where the source explicitly records the year
Time standard: UTC, subject to the Linux log's missing year

1. The question

Did the Windows and Linux logs show suspicious authentication activity, what actions followed the successful logins, and what can the available evidence establish about the source devices and possible connection between the events?

2. Evidence table

The table below separates the evidence itself from the interpretation. Text in the Exact evidence column is quoted from the log transcription.

Evidence item and source	Recorded time (UTC)	Exact evidence	What it shows
Windows Event 4625 — window.log	14:02:11	2026-03-14T14:02:11Z EventID=4625 Computer=SRV-FS01 TargetUserName=jdoe LogonType=3 SubStatus=0xC000006A IpAddress=10.20.5.60 WorkstationName=WS-0207	A failed network logon for jdoe to SRV-FS01 from 10.20.5.60.
Windows Event 4625 — window.log	14:02:13	2026-03-14T14:02:13Z EventID=4625 Computer=SRV-FS01 TargetUserName=mrivera LogonType=3 SubStatus=0xC000006A IpAddress=10.20.5.60 WorkstationName=WS-0207	A failed network logon for a second account from the same source.
Windows Event 4625 — window.log	14:02:15	2026-03-14T14:02:15Z EventID=4625 Computer=SRV-FS01 TargetUserName=asmith LogonType=3 SubStatus=0xC000006A IpAddress=10.20.5.60 WorkstationName=WS-0207	A failed network logon for a third account from the same source.
Windows Event 4625 — window.log	14:02:17	2026-03-14T14:02:17Z EventID=4625 Computer=SRV-FS01 TargetUserName=svc-backup LogonType=3 SubStatus=0xC000006A IpAddress=10.20.5.60 WorkstationName=WS-02	A failed network logon for svc-backup. The workstation field is incomplete in the supplied transcription.

Windows Event 4625 — window.log	14:02:19	2026-03-14T14:02:19Z EventID=4625 Computer=SRV-FS01 TargetUserName=admin LogonType=3 SubStatus=0xC0000064 IpAddress=10.20.5.60 WorkstationName=WS-0207	A failed network logon using a username that the status code indicates was not found.
Windows Event 4625 — window.log	14:02:21	2026-03-14T14:02:21Z EventID=4625 Computer=SRV-FS01 TargetUserName=helpdesk LogonType=3 SubStatus=0xC0000064 IpAddress=10.20.5.60 WorkstationName=WS-0207	Another failed network logon using a username that was not found.
Windows Event 4624 — window.log	14:02:24	EventID=4624 Computer=SRV-FS01 TargetUserName=svc-backup LogonType=3 AuthenticationPackageName=NTLM IpAddress=10.20.5.60	A successful network logon for svc-backup from the same source IP, three seconds after the last recorded failure.
Windows Event 4672 — window.log	14:02:24	2026-03-14T14:02:24Z EventID=4672 Computer=SRV-FS01 SubjectUserName=svc-backup PrivilegeList=SeBackupPrivilege, SeRestorePrivilege LogonId=0x5f1a2	Special privileges, including backup and restore privileges, were assigned to the svc-backup logon.
DHCP lease table — dhcp.log	14:00	DHCP lease table at 2026-03-14 14:00 UTC	The DHCP information is a snapshot recorded before the Windows activity.
DHCP device description — dhcp.log	14:00	(shared kiosk in the warehouse, no regular user)	The supplied DHCP information describes one device as a shared warehouse kiosk without a regular user. The IP-to-device mapping should be confirmed against the original lease table before treating it as definitive.
Windows Event 4720 — window.log	14:05:40	2026-03-14T14:05:40Z EventID=4720 Computer=SRV-FS01 TargetUserName=helpdesk2 SubjectUserName=svc-backup	The svc-backup account is recorded as the actor associated with creation of helpdesk2.
Windows Event 4732 — window.log	14:05:52	2026-03-14T14:05:52Z EventID=4732 Computer=SRV-FS01 MemberName=helpdesk2 TargetGroup=Administrators SubjectUserName=svc-backup	The helpdesk2 account was added to the Administrators group by svc-backup.
Windows Event 4624 — window.log	14:06:10	2026-03-14T14:06:10Z EventID=4624 Computer=SRV-FS01 TargetUserName=helpdesk2 LogonType=10 IpAddress=10.20.5.60 WorkstationName=WS-0207 LogonId=0x5f6c0	A successful RemoteInteractive logon for helpdesk2 from the same source IP. Type 10 is commonly associated with Remote Desktop.

Linux failed SSH login — auth. log	14:10:01	Mar 14 14:10:01 WEB-01 sshd[2201]: Failed password for invalid user admin from 198.51.100.77 port 41022 ssh2	A failed SSH password attempt using an invalid username.
Linux failed SSH login — auth. log	14:10:03–14:10:11	Failed password entries for test, root, root, ubuntu, and deploy from 198.51.100.77	Five more failed password attempts from the same source. The two root attempts are recorded without the invalid user wording; deploy is also recorded without it.
Linux successful SSH login — auth. log	14:11:30	Mar 14 14:11:30 WEB-01 sshd[2230]: Accepted password for deploy from 198.51.100.77 port 41110 ssh2	A successful password-based SSH login for deploy from the same source as the earlier failures.
Linux session record — auth. log	14:11:30	Mar 14 14:11:30 WEB-01 sshd[2230]: pam_unix(sshd:session): session opened for user deploy(uid=1001)	A session was opened for deploy, UID 1001.
Linux sudo record — auth. log	14:12:30	Mar 14 14:12:30 WEB-01 sudo: deploy : TTY=pts/0 ; PWD=/home/deploy ; USER=root ; COMMAND=/usr/bin/cat /etc/shadow	The deploy account used sudo to run cat /etc/shadow as root. The log does not show whether the file contents were copied or transmitted elsewhere.
Linux account creation — auth. log	14:13:02	Mar 14 14:13:02 WEB-01 useradd[2301]: new user: name=support, UID=1002, GID=1002	A new Linux account named support was created. This line alone does not establish who initiated the command.
Linux public-key login — auth. log	14:20:15	Mar 14 14:20:15 WEB-01 sshd[2410]: Accepted publickey for asmith from 10.20.5.15 port 52044 ssh2: RSA SHA256:Qx9dA7vN2uXkq1sB0c1Zz0A	A public-key SSH login for asmith was accepted from 10.20.5.15. The DHCP information supplied describes an admin jump host used by asmith, but the IP-to-host mapping needs confirmation.

Evidence handling note: The Windows successful-login record at 14:02:24 was split or damaged in the supplied transcription around the Workstation and LogonId fields. I have quoted only the intact portion. The Linux log header says it is trimmed, has no year or time zone in the file, and that the host clock is UTC. Therefore, its March 14 timestamps can be treated as UTC according to that header, but the year cannot be established from that file alone.

3. Analysis of the evidence

A. Windows authentication activity

Between 14:02:11 and 14:02:21, six failed network logons were recorded against different usernames from the same IP address, 10.20.5.60.

The status codes are important:

- 0xC000006A indicates that the username was recognized but the password was incorrect.
- 0xC0000064 indicates that the username does not exist.

Four attempts used usernames associated with existing accounts: jdoe, mrivera, asmith, and svc-backup. Two attempts used usernames that were not found: admin and helpdesk.

Assessment: This pattern is consistent with possible password spraying or username enumeration. It is not enough to prove password spraying because the logs do not reveal whether the same password was tried against each account. It is also not enough to establish credential stuffing, because there is no evidence that stolen username-and-password pairs were used.

Three seconds after the last recorded failure, svc-backup successfully logged in using NTLM. The Event 4672 record then shows that the account received SeBackupPrivilege and SeRestorePrivilege. Later, the same account was recorded as creating helpdesk2 and adding it to the Administrators group. At 14:06:10, helpdesk2 successfully logged in using Logon Type 10.

Assessment: The sequence is suspicious and may indicate misuse or compromise of the svc-backup account. However, the event records do not independently prove that the account was compromised or that the actions were unauthorized.

B. Linux authentication activity

The Linux log records six failed SSH password attempts from 198.51.100.77 between 14:10:01 and 14:10:11.

The attempted usernames were:

Username	What the log indicates
admin	Invalid user
test	Invalid user
root	No invalid-user wording; username appears to be recognized
ubuntu	Invalid user
deploy	No invalid-user wording; username appears to be recognized

The root attempt occurred twice.

At 14:11:30, the log records a successful password login for deploy from the same IP address. This was 79 seconds after the last failed attempt at 14:10:11.
After the successful login:

1. A session opened for deploy.
2. At 14:12:30, deploy used sudo to read /etc/shadow.
3. At 14:13:02, a new account named support was recorded as created.

Assessment: The sequence is suspicious because failed authentication attempts were followed by a successful login and activity involving a sensitive system file and a new user account. The log does not establish whether the successful login was unauthorized, whether the contents of /etc/shadow were taken, or who executed the account-creation command.

C. Are the Windows and Linux events connected?

The first recorded Windows failure occurred at 14:02:11. The first Linux failure occurred at 14:10:01. The difference is **7 minutes and 50 seconds**.

Detail	Windows	Linux
Target	SRV-FS01	WEB-01
Source IP	10.20.5.60	198.51.100.77
First recorded failure	14:02:11 UTC	14:10:01, host clock UTC
Later activity	Privileged account creation and remote logon	Successful SSH login, sudo command and account creation

What is established: Both systems recorded suspicious authentication-related activity on March 14. The events began approximately eight minutes apart and have different recorded source IP addresses.

What is not established: The logs do not prove that the same attacker, person, or device was responsible for both sets of activity.

Connecting hypothesis: One operator may have been attempting to access multiple systems in the environment.

Non-connecting hypothesis: The Windows and Linux events may have been independent activities carried out by different people or processes.
Both hypotheses remain unproven.

D. The asmith public-key login

At 14:20:15, the Linux log records an accepted public-key login for asmith from 10.20.5.15.
The DHCP information you supplied describes an admin jump host used by asmith. If the mapping between 10.20.5.15 and JUMP-01 is confirmed, this login may be consistent with expected administrative access.
Public-key authentication is not inherently suspicious. I would check:

- Whether 10.20.5.15 is confirmed as JUMP-01.
- Whether asmith was expected to log in at that time.
- Whether the RSA key fingerprint belongs to the authorized user or system.
- Whether the login appears in approved administrative activity or change records.

I would not classify this login as malicious based only on the supplied evidence.

4. What the evidence could not tell me

The available evidence has several limitations:

1. **Identity of the Windows operator**: The DHCP description identifies the suspected source device as a shared warehouse kiosk, if the proposed mapping is correct. It does not identify the person using it.
2. **Identity of the Linux operator**: The Linux log records the source IP address but does not identify the person controlling that source.
3. **Whether the Windows activity was authorized**: The logs do not include a change ticket, maintenance record, or approval for the account creation and group modification.
4. **Whether the password was sprayed**: The Windows records show multiple accounts being targeted, but they do not show the actual passwords attempted.
5. **Whether Linux credentials were compromised**: The successful deploy login followed failed attempts, but the log does not establish how the credentials were obtained.
6. **Whether /etc/shadow was exfiltrated**: The sudo log records the command, not what happened to the output.
7. **Who created support**: The supplied useradd line does not identify the initiating user.
8. Whether both platforms were attacked by the same operator: There is no direct evidence connecting the two source IP addresses or the activity.
9. **Whether the asmith login was legitimate**: The DHCP description suggests a possible administrative jump host, but the mapping and authorization need verification.
10. **Exact Linux event year**: The Linux log does not include a year, so it cannot independently establish that year as 2026.

5. One decision I would make differently now

If I were investigating this activity again, I would prioritize verifying the identity and activity of the source device WS-0207 before making conclusions about the person responsible.
I would request the endpoint logon records, physical access records, and network access information for the shared warehouse kiosk. I would also preserve the relevant Windows and Linux logs and investigate the svc-backup and deploy accounts separately.
This would help distinguish suspicious activity from authorized administration and avoid attributing actions to a person without supporting evidence.

6. Confidence assessment

Medium confidence

I have medium confidence that the logs show suspicious authentication and account-management activity.

Reasons for confidence:

- Multiple failed Windows logons targeted different accounts from one source.
- A successful svc-backup login followed shortly after the failures.
- The svc-backup account was associated with creating an account and adding it to Administrators.
- Linux failed SSH attempts were followed by a successful deploy login.
- The Linux session was followed by access to /etc/shadow and account creation.

What would raise my confidence:

- Original, complete Windows and Linux log files.
- Confirmed DHCP IP-to-hostname mappings.
- Endpoint detection and response (EDR) records.
- Network and firewall logs.
- Authentication server records.
- Approved change tickets and administrative activity records.
- Records identifying who used the shared kiosk.

What would lower my confidence:

- Evidence that the account creation and group modification were approved.
- Evidence that the logins were part of scheduled maintenance.
- Confirmation that the source devices and accounts were being used by authorized administrators.
- Evidence that the apparent sequence resulted from legitimate automated processes.

My confidence in the **suspicious pattern** is higher than my confidence in identifying the person responsible or proving that the Windows and Linux events were connected.